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Fourth Semester B.E./B.Tech. Degree Examination, Dec.2025/Jan.2026 Discrete Mathematical Structures

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Define a logically equivalent. Show that $[(p \vee q) \rightarrow r] \leftrightarrow [\neg r \rightarrow \neg (p \vee q)]$ for primitive statements p, q, r is logically equivalence.	6	L1	CO1
	b.	Establish the validity of the following argument using the rules of inference: $\{(p \rightarrow q) \wedge (r \rightarrow s) \wedge (\neg q \vee \neg s)\} \rightarrow \neg (p \wedge r)$	7	L2	CO1
	c.	i) Give an indirect proof of the statement: “The product of two even integers is an even integer”. ii) Provide a proof by contradiction of the following statement: “For every integer n, if n^2 is odd, then n is odd”.	7	L3	CO1
OR					
Q.2	a.	Define a tautology. Prove that, for an propositions p, q, r the compound proposition. $\{p \rightarrow (q \rightarrow r)\} \rightarrow \{(p \rightarrow q) \rightarrow (p \rightarrow r)\}$ is tautology.	6	L1	CO1
	b.	Using law of logic, prove the result. $[(p \vee q) \wedge (p \vee \neg q)] \vee q \Leftrightarrow p \vee q$	7	L2	CO1
	c.	Find whether the following argument is valid: If a triangle has two equal sides, then it is isocles. If a triangle is isocles, then it has two equal angles. <u>A certain triangle PQR does not have two equal angles.</u> $\therefore$ The triangle PQR does not have two equal sides.	7	L3	CO1
Module – 2					
Q.3	a.	By mathematical induction, prove that $(n!) \geq 2^{n-1}$ for all integers $n \geq 1$.	6	L2	CO2
	b.	Find an in explicit form for the following: i) $a_1 = 4, a_n = a_{n-1} + n$ for $n \geq 2$ ii) $a_1 = 7, a_n = 2a_{n-1} + n$ for $n \geq 2$	7	L2	CO2
	c.	Find how many distinct four-digit integers one can make from the digits 1, 3, 3, 7, 7, 8	7	L3	CO2

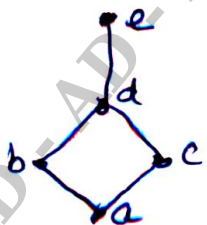
OR

Q.4	a.	Prove that $4n < (n^2 - 7)$ for all positive integers $n \geq 6$.	6	L2	CO2
	b.	i) How many arrangements are there for all letters in the word SOCIOLOGICAL? ii) In how many of these arrangements all the vowels are adjacent?	7	L3	CO2
	c.	Determine the coefficient of i) xyz^2 in the expansion of $(2x - y - z)^4$ ii) $x^2y^2z^3$ in the expansion of $(3x - 2y - 4z)^7$	7	L2	CO2

Module – 3

Q.5	a.	Let $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{6, 7, 8, 9, 10\}$. If a function $f : A \rightarrow B$ is defined by $f = \{(1, 7), (2, 7), (3, 8), (4, 6), (5, 9), (6, 9)\}$ determine $f^{-1}(6)$ and $f^{-1}(9)$. If $B_1 = \{7, 8\}$ and $B_2 = \{8, 9, 10\}$, find $f^{-1}(B_1)$ and $f^{-1}(B_2)$.	6	L2	CO2
	b.	State Pigeon-hole principle. ABC is an equilateral triangle whose sides are of length 1cm each. If we select 5 points inside the triangle, prove that atleast two of these points are such the distance between them is less than $1/2$ cm.	7	L3	CO2
	c.	Let f and g be functions from R to R defined by $f(x) = ax + b$ and $g(x) = 1 - x + x^2$ if $(gof)(x) = 9x^2 - 9x + 3$, determine a & b .	7	L2	CO2

OR

Q.6	a.	Let f, g, h be functions from z to z defined by $f(x) = x - 1$, $g(x) = 3x$, $h(x) = \begin{cases} 0, & \text{if } x \text{ is even} \\ 1, & \text{if } x \text{ is odd} \end{cases}$ Determine $(f \circ (g \circ h))(x)$ and $((f \circ g) \circ h)(x)$ and verify that $f \circ (g \circ h) = (f \circ g) \circ h$.	6	L2	CO3
	b.	For $A = \{a, b, c, d, e\}$, the Hasse diagram for the poset (A, R) is as shown in Fig.Q.6(b). i) Determine the relation matrix for R . ii) Construct the diagram for R .  Fig.Q.6(b)	7	L3	CO3
	c.	Let $A = \{1, 2, 3, 4\}$ and let R be the relation on A defined by xRy if and only if " x divides y ", i) Write down R as a set of ordered pairs ii) Draw the diagram of R iii) Determine the in-degrees and out-degrees of the vertices in the digraph.	7	L3	CO3

Module – 4

Q.7	a.	In how many ways 5 number of a's, 4 number of b's and 3 number of c's can be arranged so that all the identical letters are not in a single block.	6	L2	CO4
	b.	Find the number of permutations of the letters a, b, c, x, y, z in which none of the patterns spin, game, path or net occurs.	7	L3	CO4
	c.	Solve the recurrence relation $a_n - 6a_{n-1} + 9a_{n-2} = 0$ for $n \geq 2$, given that $a_0 = 5, a_1 = 12$.	7	L2	CO4

OR

Q.8	a.	Find the number of derangements of 1, 2, 3, 4. Mention them.	6	L2	CO4
	b.	By using the expansion formula, find the rook polynomial for the board c shown below (madeup of unshaded parts) : 	7	L3	CO4
	c.	Solve the recurrence relation $a_n - 3a_{n-1} = 5 \times 3^n$ for $n \geq 1$ given that $a_0 = 2$	7	L2	CO4

Module – 5

Q.9	a.	Let $w_4 = \{1, -1, i, -i\}$, the set of all fourth roots of unity. Show that w_4 is an abelian group under x.	6	L2	CO5
	b.	Define Klein-4-group. If $A = \{e, a, b, c\}$ then show that this is a Klein-4 group.	7	L2	CO5
	c.	State and prove Lagrange's theorem.	7	L2	CO5

OR

Q.10	a.	Show that the group $(G, *)$ whose multiplication table is as given below is cyclic. <table><tr><td>*</td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td></tr><tr><td>a</td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td></tr><tr><td>b</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td><td>a</td></tr><tr><td>c</td><td>c</td><td>d</td><td>e</td><td>f</td><td>a</td><td>b</td></tr><tr><td>d</td><td>d</td><td>e</td><td>f</td><td>a</td><td>b</td><td>c</td></tr><tr><td>e</td><td>e</td><td>f</td><td>a</td><td>b</td><td>c</td><td>d</td></tr><tr><td>f</td><td>f</td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td></tr></table>	*	a	b	c	d	e	f	a	a	b	c	d	e	f	b	b	c	d	e	f	a	c	c	d	e	f	a	b	d	d	e	f	a	b	c	e	e	f	a	b	c	d	f	f	a	b	c	d	e	6	L	CO
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	b.	Let $G = S_4$, the symmetric group of degree 4. For $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$. Find the subgroup $H = \langle \alpha \rangle$. Also, determine the number of left cosets of H in G .	7	L3	CO5																																																	
	c.	If $*$ is an operation on z defined by $x * y = x + y + 1$, prove that $(z, *)$ is an abelian group.	7	L2	CO5																																																	
